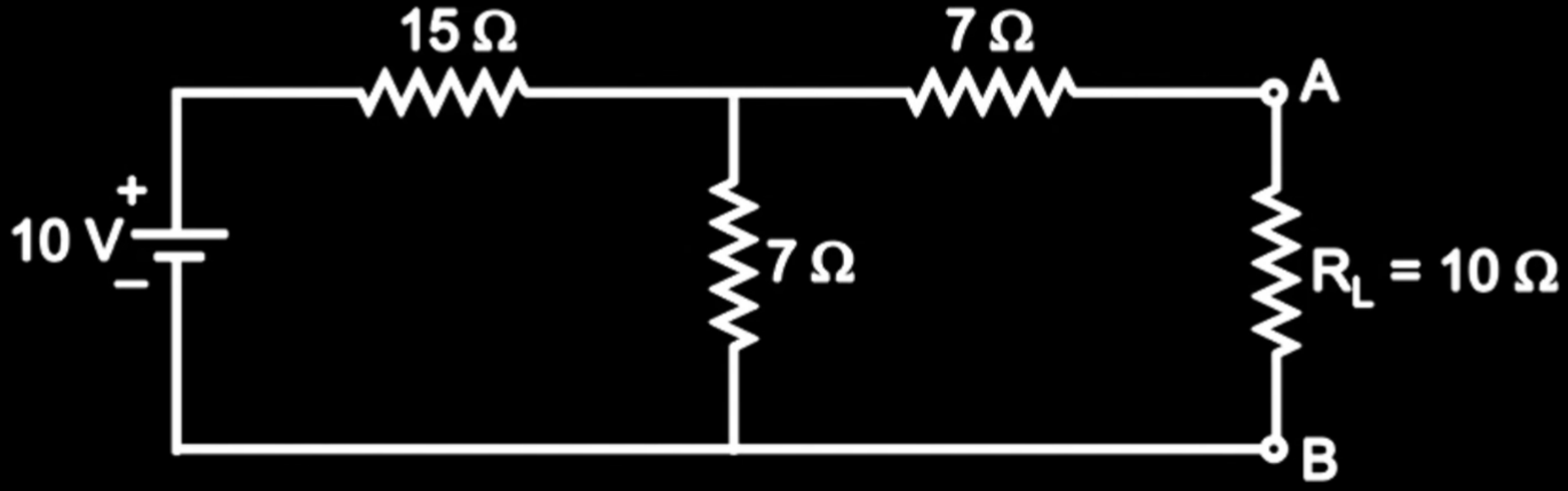
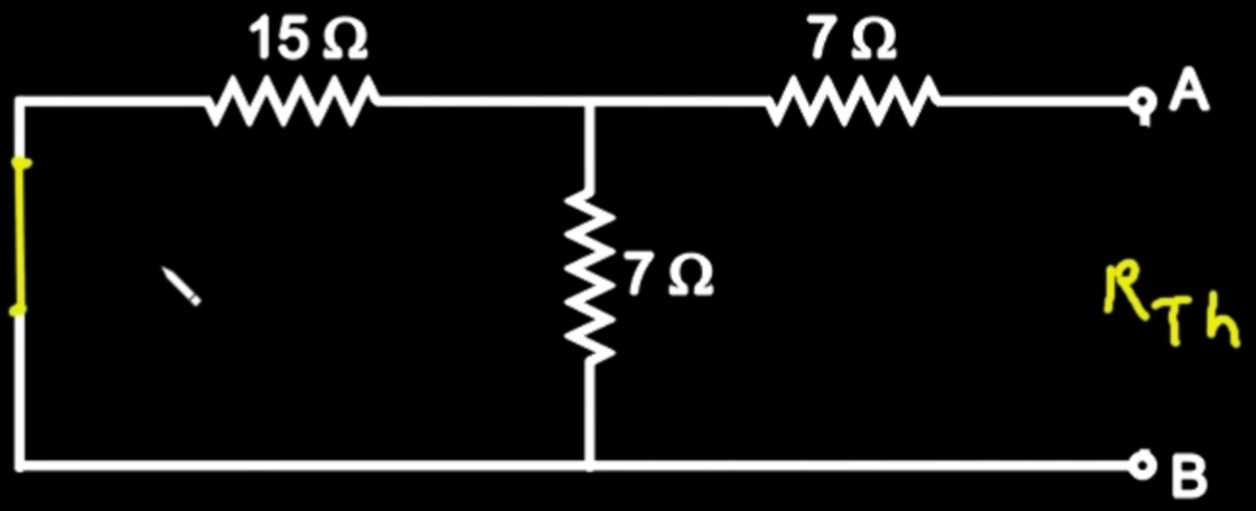
THEVENIN'S THEOREM (https://electrical-engineering.app/)

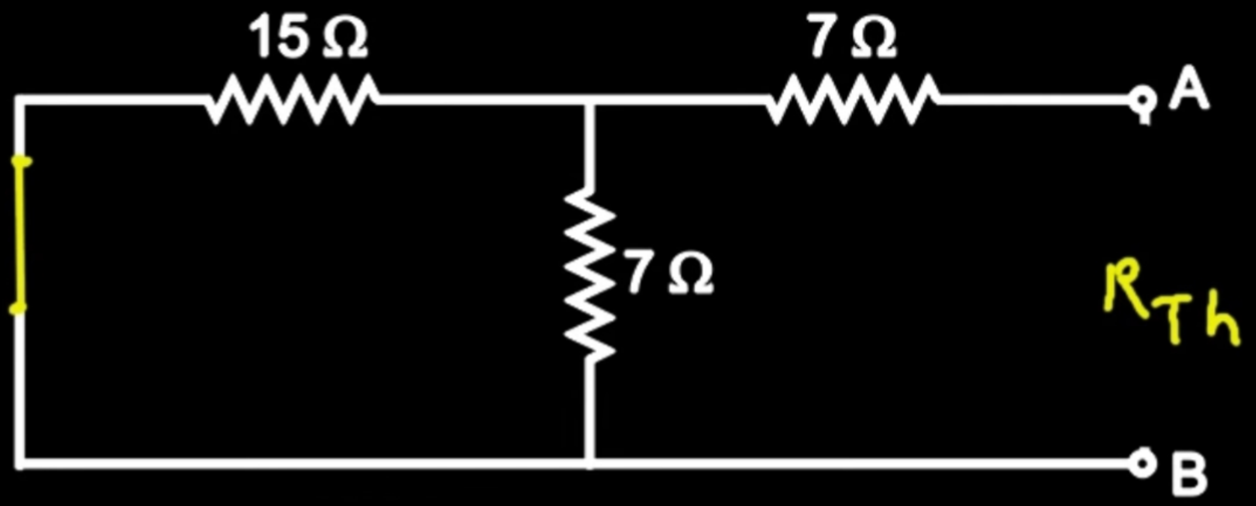
Q. Develop Thevenin's equivalent circuit between the points A and B in Fig. and find current in RL = 10 Ohm.



Solution :

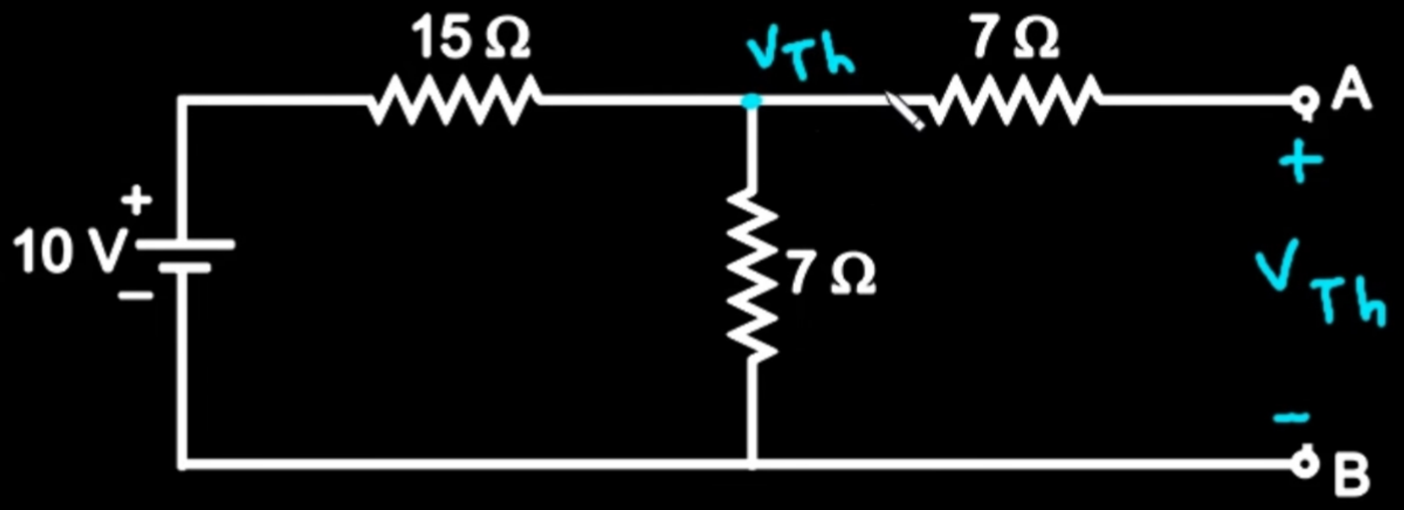
① To find R_{Th}, Remove R_L & turn off all independent sources

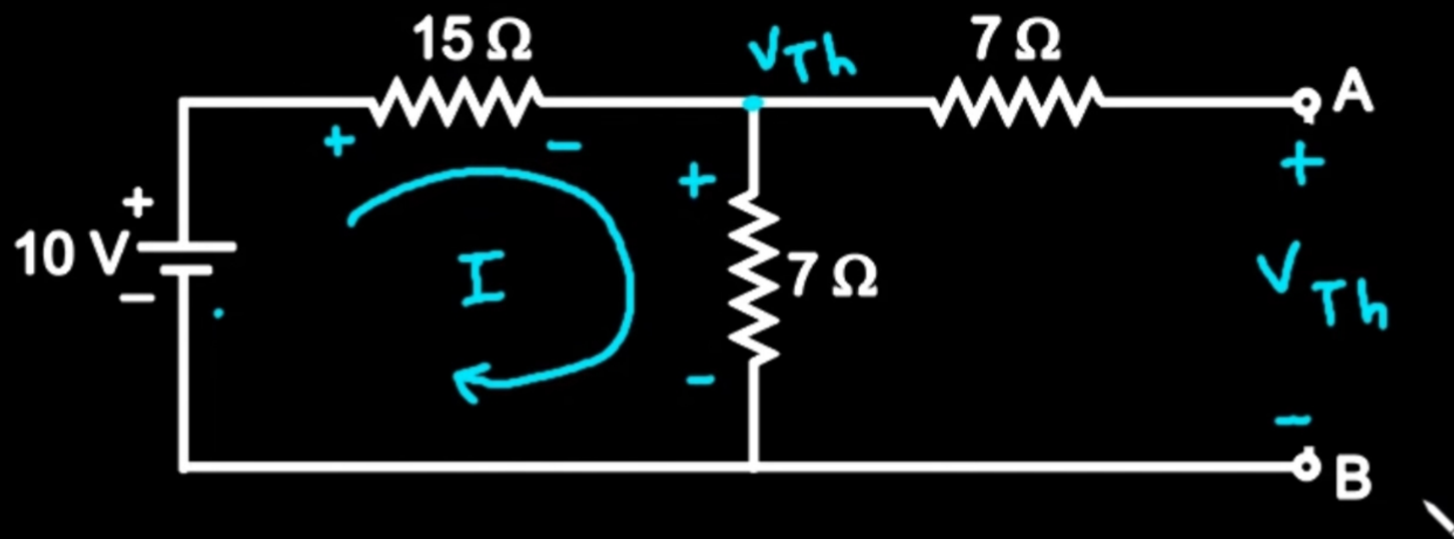




$$R_{Th} = 7 + \frac{15 \times 7}{15 + 7} = \frac{259}{22}\,\Omega$$

② To find V_{Th}, Remove R_L & find V_{oc}





Apply KVL to loop

$$-10 + 15I + 7I = 0$$

$$-10 + 22I = 0$$

$$22I = 10$$

$$I = \frac{10}{22}\ A$$

$$V_{Th} = 7 \times I = 7 \times \frac{10}{22} = \frac{35}{11} V$$

$$\boxed{V_{Th} = 3.181 \text{ V}}$$

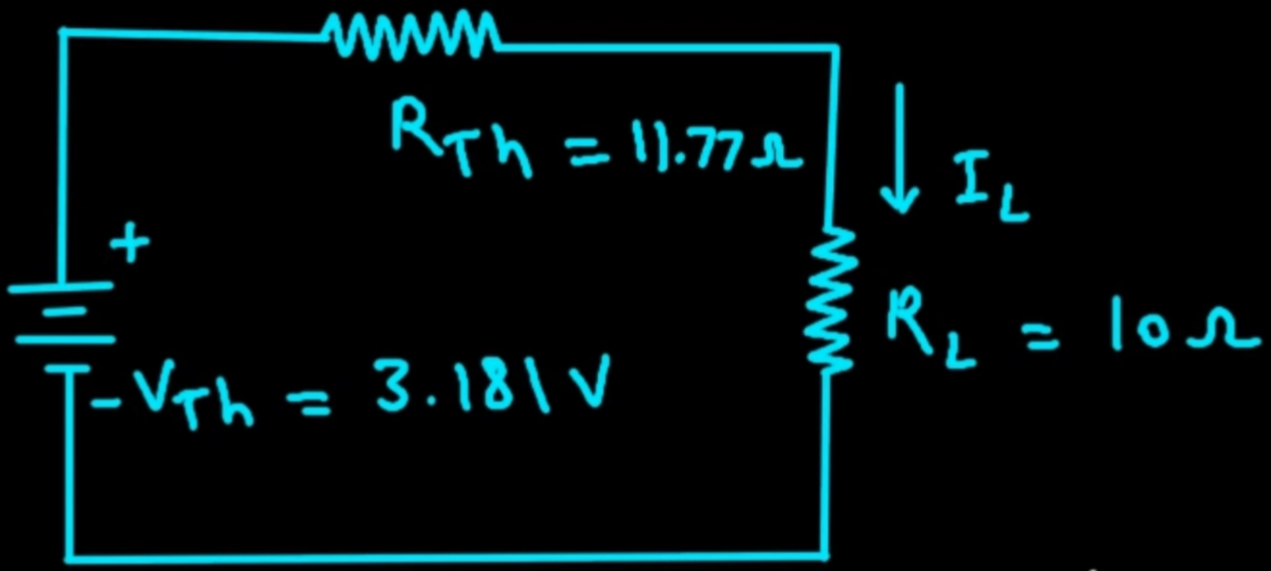


fig. Thevenin's equivalent circuit

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{3.181}{11.77 + 10} = 0.146 \text{ A}$$